

## RESEARCH ARTICLE

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# Effects of grazing exclusion on soil infiltrability and preferential flow in savannah livestock grazing systems

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## Abstract

Although the establishment of grazing exclosures is often considered an important strategy of reducing soil degradation, their effect on hydrological functions is still poorly understood, particularly in dry-hot savanna systems with heavy livestock grazing. In addition, the different roles played by the herb roots and the niche differentiation of woody plant roots induced by the grazing and exclosure in soil hydrological functions are still unclear. This study aimed to evaluate the effect of exclosure management on hydrological functions and services and assess the role of different functional vegetation types in improving these functions and services. The experiments were conducted in the continuous grazing areas and exclosure areas in a savanna ecosystem in Yuanjiang, southwest China. The root biomass of herb and woody plants, soil physical properties, water infiltration, and preferential water flow were measured in each study site. The results showed a significant improvement in soil physical properties with more termite holes, lower soil bulk density, and higher total and non-capillary soil porosity in exclosure areas compared with grazing ones. As a result, the initial and average water infiltration, and preferential flow significantly increased by 132.76%, 185.31%, and 30.29%, respectively, in exclosure areas compared with the grazing ones. The results also showed that the exclosure significantly improved the soil surface water content by increasing the root biomass of herb and woody plants in the shallow soil layers. Herb plants probably improved water infiltration and insulated evaporation, while woody plants improved lateral preferential water flow paths and accelerated the water exchange between vertical and horizontal soil layers, thereby improving the water availability for different plants. This study demonstrated that the exclosure-induced restoration and niche differentiation of herb and woody plants, coupled with improved soil structure in exclosures, could promote hydrological function and services and improve water management in savanna systems. Therefore, the grazing exclosures could be expanded periodically and cautiously to prevent soil degradation in the present savanna systems.

## KEYWORDS

exclosure management, hydrological function, preferential flow, savanna system, soil infiltration

## 1 | INTRODUCTION

Soil degradation, caused by over-grazing, erosion, water depletion, excessive cultivation, and fire, is a serious problem that will continue to aggravate in savanna ecosystems globally in the coming decades (Quaranta et al., 2020; Zhou et al., 2010). The soil of the savanna ecosystem of Yuanjiang China is seriously degraded, especially owing to livestock grazing. Grazing-based animal activities have numerous negative environmental consequences, such as biodiversity reduction (Savadojo et al., 2017), compacted soil structure (Zhang et al., 2019), decreased soil fertility (Liu et al., 2016; Pulido et al., 2018), excessive soil erosion (Wine et al., 2012), and impaired hydrological characteristics (Hoshino et al., 2009). In response to these problems, the Chinese Government and scientists have proposed the establishment of exclosures, which is widely regarded as an efficient, effective, and inexpensive practice for the restoration of degraded soils (Feyisa et al., 2017; Tóth et al., 2018). Previous research has demonstrated that the use of exclosures in the present savanna region can help restore the vegetation cover (Jin et al., 2017), while their use in other regions helps prevent soil erosion and improves soil nutrient levels (Fernández-Lugo et al., 2009; Feyisa et al., 2017; Koci et al., 2020). However, although exclosures are considered useful in mitigating soil degradation (Angassa, 2014; Koci et al., 2020; Liu et al., 2016), their effects on ecohydrological ecosystem services (e.g. water infiltration and distribution, water flow patterns, and water supply regulation) in the savanna system was unknown. In addition, evidence on the effect of plant species (herb and woody plants) on soil hydrological functions is basically absent in savanna systems.

The soil water movement, a constituent of water cycle, is driven by the complex water transfer process within the soil–plant–atmosphere system that is connected to climatic and environmental fluctuations, including rainfall, evapotranspiration, and water coverage (Daly & Porporato, 2006). As water is the limiting factor for the growth and productivity of terrestrial ecosystems, especially in the Yuanjiang savanna ecosystem with high evapotranspiration (2750 mm per year) and low rainfall (786.6 mm yr<sup>-1</sup>), it requires more attention regarding the appropriate land use and pasture management to improve soil water status. Livestock grazing affects watershed hydrology mainly by altering soil properties and vegetation cover (Vandendorj et al., 2017). Additionally, livestock grazing can affect hydrologic processes directly by increasing the soil bulk density and reducing soil porosity and water infiltration through the downward pressure imposed on the soil surface by grazing animals, thereby resulting in low water input from rainfall into deep soil layers (Sharro, 2007). Conversely, non-grazing can positively influence water infiltration, flow, and redistribution by promoting the restoration of above- and underground plant biomass (Estragnat et al., 2018; Sharro, 2007; Teague et al., 2011). In addition, macropores formed by root systems, animal burrows, and cracks can promote the formation of preferential water flow pathways in each soil layer, which results in the redistribution of soil water.

Plants can influence climatological and hydrological processes, such as rainfall distribution, evaporation, runoff generation,

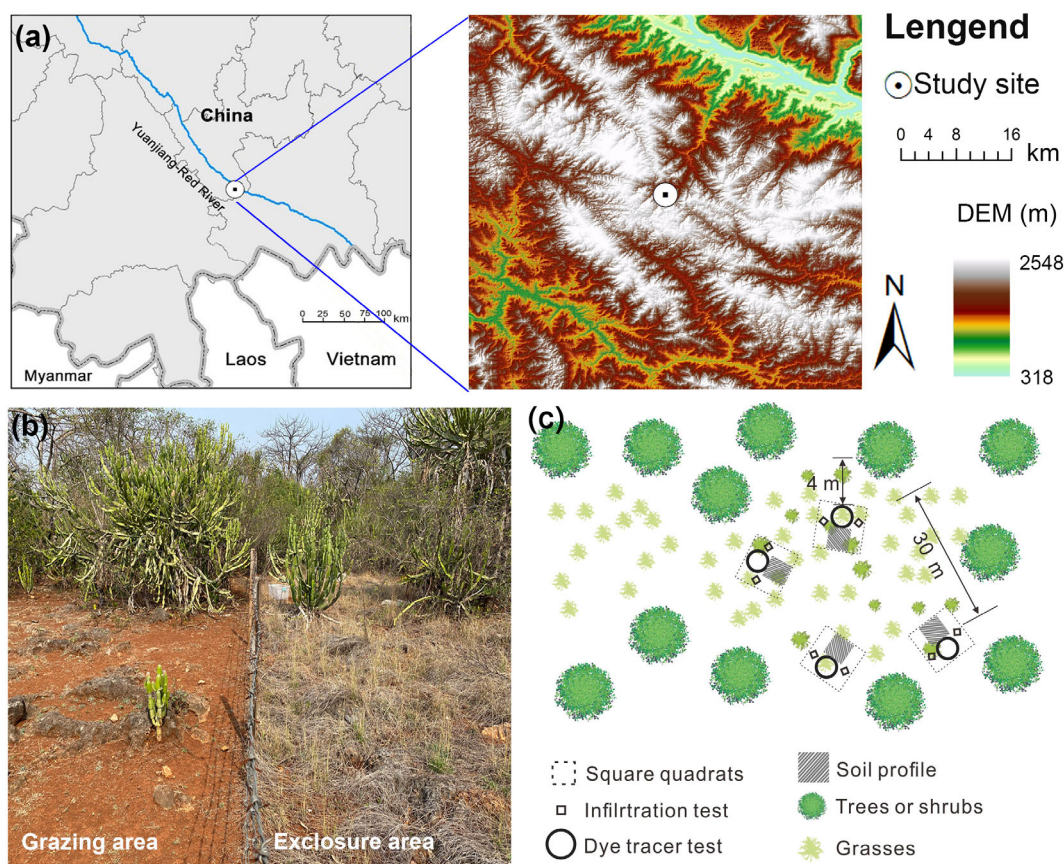
infiltration, preferential and lateral flow, and water exchange, which modify the water balance (Benegas et al., 2014; Zhu et al., 2019). Alternating between grazing and non-grazing can promote vegetation persistence and differentiation, especially in terms of the biomass and species diversity of herbs (Su et al., 2018). The functional vegetation of savannas can be divided into woody species (e.g. trees and shrubs) and herbs (grasses). However, various functional plant communities have different effects on hydrological processes. A previous study found that although the encroachment of open woodlands by shrubs increases soil infiltration, the infiltration in the shrubland interspaces significantly decreases (1.90-fold) compared with that in grasslands, thereby suggesting a higher contribution of herbs than of trees and shrubs to soil infiltration (Eldridge et al., 2015), while it inevitably influences water availability. Herb and woody plants exhibit rooting and growth differentiation, and they predominantly respond to soil water availability at depths of 5 cm and 40 cm, respectively (Holdo & Nippert, 2015). However, unlike herbs, woody plants are highly plastic in terms of their water uptake patterns, as they absorb shallow soil water in the rainy season and deep soil water (below 30 cm in depth) in the dry season (Holdo & Nippert, 2015). Thus, grazing- and exclosure-induced changes in the herb root biomass and the niche differentiation of woody plant roots in each soil layer may play different, yet still unclear, roles in soil hydrological functions.

In order to unravel the effect of exclosures on soil hydrological functions and services and the contribution of different types of functional vegetation, this study investigated soil infiltrability and water flow behaviours in the grazing and exclosure plots of a savanna ecosystem. Our study aimed to assess the effect of grazing exclusion on water storage, soil infiltrability, and preferential water flows and explore related influencing factors in dry-hot savanna systems (southwest China). We hypothesised that: (1) the soil infiltrability and preferential water flow would increase in the exclosure area; (2) the improved hydrological functions are closely linked to the soil structure and vegetation restoration, and; (3) herbs tend to increase water infiltration, while the woody plants especially improve lateral preferential water flow paths in savanna systems.

## 2 | MATERIALS AND METHODS

### 2.1 | Study site and experimental design

The study site was located in the Yuanjiang Savanna Ecosystem Research Station (YSERS; 23°27'N, 102°10'E) in Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, in southwestern China. The climate of the area is dry and hot, and the average annual temperature is 24.0°C. The total mean annual precipitation is 786.6 mm, with approximately 80.0% of the annual rainfall occurring during the rainy season (from May to October). The total sunshine duration is 2261.7 hr, and the average pan evapotranspiration is 2750 mm yr<sup>-1</sup> (Li et al., 2018). The dominant tree, shrub, and herb species in the region are *Lannea coromandelica* and *Polyalthia cerasoides*, *Strophoblachia fimbriicalyx*, *Woodfordia fruticosa*, and



**FIGURE 1** Location of (a) the study site in Yunnan Province, southwestern China, (b) the grazing area and the enclosure area, and (c) the sketch maps of experimental design and field samples measurement in enclosure area. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4388)]

**TABLE 1** Soil physicochemical properties in grazing area and enclosure area

| Treatment      | pH                 | SOC ( $\text{g kg}^{-1}$ ) | Total N ( $\text{g kg}^{-1}$ ) | Sand (%)           | Silt (%)            | Clay (%)            |
|----------------|--------------------|----------------------------|--------------------------------|--------------------|---------------------|---------------------|
| Grazing area   | $6.72 \pm 0.22$    | $2.48 \pm 0.15$            | $26.19 \pm 0.22$               | $23.79 \pm 0.72$   | $52.28 \pm 1.14$    | $23.93 \pm 0.52$    |
| Enclosure area | $6.17 \pm 0.16$ ns | $2.86 \pm 0.12$ ns         | $31.73 \pm 0.27^{**}$          | $25.5 \pm 0.46$ ns | $49.82 \pm 0.77$ ns | $24.68 \pm 0.67$ ns |

Note: ns, no significant difference.

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

*Heteropogon contortus*, *Arthraxon lanceolatus*, and *Euphorbia royleana*, respectively (Jin et al., 2017).

The research area was continuously grazed with local livestock of 90% cattle and 10% goats. The grazing intensity in the grazing area was 0.3–0.5 stock per ha because of limited food. The typical live weights of cattle and goats were 213–462 kg and 41–47 kg, respectively. The grazing and enclosure areas were separated using barbed wire (Figure 1). The enclosure system (total 10 ha) was established in March 2012 and its use spanned a period of more than 8 years. The enclosure and grazing areas had common characteristics (parent material, slope degree and direction, geographical position and altitude). The primary soil physicochemical properties in grazing and enclosure areas are given in Table 1. The herbs grew naturally and were not affected by grazing and browsing disturbances in the enclosure area. However, the grazing area presented a bare landscape with few herbs on the surface owing to long-term heavy grazing. Compared with the grazing area, the enclosure area had higher biomass of shrub plants,

but the woody plant biomass did not differ between both areas because livestock rarely consumed woody plants (Jin et al., 2017). Four plots ( $3.0 \times 3.0$  m) were established in the grazing and enclosure areas (a total of eight plots), respectively, with a minimum distance between two plots of 30 m. The sketch map of experimental design and field samples measurement for the water infiltration, dye tracing experiment, plant root, and soil sampling are given in Figure 1.

## 2.2 | Dye tracer infiltration experiment

To determine the water infiltration and to visualise water flow pathways, eight dye tracing experiments were conducted in the grazing and enclosure areas in April 2020 (Figure 1). The infiltration experiment was conducted in the grass carpet, which was approximately 4 m from the nearest trees (Figure 1c). However, some areas of the study site were covered with stones both above- and belowground,

which could influence the results of water infiltration and water flow path experiments. During the experiments, the large stone areas were excluded to obtain a relatively complete and representative vertical soil profile. A hollow stainless steel cylinder (diameter: 20 cm; height: 30 cm) was inserted 5 cm into the soil while ensuring that the side-walls and edges were watertight. Then, tapwater (25 L in total) dyed with brilliant blue FCF at a concentration of 4.0 g L<sup>-1</sup> was gradually poured into the cylinder at selected time intervals following the infiltration measurement procedure. The initial infiltration rates were calculated on the basis of first 3 min and the corresponding infiltration volume. The average infiltration rates were defined as the ratio of infiltration volume to the total infiltration time (Niu et al., 2019). A 60 cm (width) × 80 cm (depth) dye-stained vertical soil section was neatly excavated from the centre of the cylinder 24 hr after the end of the infiltration, using a spatula. Calibrated steel tape was placed along the horizontal and vertical lines of the soil profile before taking photographs for image analysis.

The image analysis was conducted using ArcGIS 10.3 (ESRI Inc., Redlands, CA, USA), following the procedures described in detail by Zhu et al. (2019). In short, the pictures were geometrically corrected and divided into dyed and non-dyed areas. Then the dyed area was further divided into two parts, the dye-stained concentrations of >2.0 g L<sup>-1</sup> and 0–2.0 g L<sup>-1</sup>, using a supervised classification based on the dye-staining intensity (Zhu et al., 2019). The concentration of brilliant blue FCF could reflect the water flow paths and soil water content derived from dyed solution (Jiang et al., 2019). The dye-stained areas with a concentration higher than 2.0 g L<sup>-1</sup> reflected the active water zone, which were dominated by preferential water flow. The dye-stained areas with dye-stained concentrations lower than 2.0 g L<sup>-1</sup> represent the water buffer zones, where water exchange between the preferential flow and soil matrix occur.

### 2.3 | Root biomass

Soil cores were collected along the transects at depths of 0–10, 10–20, and 20–30 cm using a 10-cm soil corer. The samples (eight replicates for each depth) were transported to the laboratory, where they were washed manually with an elutriator to collect the roots. The roots of herb and woody plants (shrubs and trees) were separated according to their morphology. Then, the fine roots of woody plants were sorted according to their physiological status (alive and dead) and diameter class (fine roots,  $d < 2$  mm; coarse roots,  $d > 2$  mm). The roots of herb and woody plants were dried at 70°C for 72 hr and then weighed.

### 2.4 | Soil physical properties

The soil physical properties were determined using bulk soil samples and employing the cutting ring method (Jiang et al., 2019). In detail, soil samples were randomly collected in four replicates at depths of 0, 15, and 30 cm with pre-weighed ( $W_{PWC}$ ) cutting cylinders (inner diameter: 50.00 mm; height: 51.00 mm; volume: 100 cm<sup>3</sup>) in the vertical soil sections. The cutting cylinders containing wet soil were

delivered to the laboratory and weighed immediately ( $W_{CRWS}$ , g). Then, we water-saturated the samples by placing them in distilled water. The water was nearly at the soil surface level, while we ensured that no amount of water entered the samples from the top of the cylinder. The water-saturated soil weight ( $W_{SAT}$ , g) was recorded after the cylinders were ponded for 24 hr. Then, the cylinders were placed on a tray with a dry sand layer on the bottom, and their weight was measured after the water in the saturated samples was drained by gravity for 2 hr and 5 d [ $W_{WD2h}$  and  $W_{WD5d}$ , respectively (g)]. The soil physical properties were determined according to the following formulae:

$$\text{Bulk density (g cm}^{-3}\text{)} = \frac{W_{CRDS}(\text{g}) - W_{HCR}(\text{g})}{100(\text{cm}^3)} \quad (1)$$

$$\text{Total porosity (\%)} = \left(1 - \frac{\text{Bulk density (g cm}^{-3}\text{)} \times 100}{\text{Soil density (g cm}^{-3}\text{)}}\right) \times 100\% \quad (2)$$

$$\text{Saturated water capacity (kg kg}^{-1}\text{)} = \frac{W_{SAT}(\text{g}) - W_{CRDS}(\text{g})}{W_{CRDS}(\text{g}) - W_{HCR}(\text{g})} \times 100\% \quad (3)$$

$$\text{Capillary porosity (\%)} = \frac{\text{Bulk density (g cm}^{-3}\text{)} \times \text{Capillary holding capacity (\%)}}{\rho_{\text{water}}(\text{g cm}^{-3})} \quad (4)$$

$$\text{Noncapillary porosity (\%)} = \text{Total porosity (\%)} - \text{Capillary porosity (\%)} \quad (5)$$

$$\text{Capillary holding capacity (\%)} = \frac{W_{WD2h}(\text{g}) - W_{CRDS}(\text{g})}{W_{CRDS}(\text{g}) - W_{HCR}(\text{g})} \times 100\% \quad (6)$$

$$\text{Field capacity (\%)} = \frac{W_{WD5d}(\text{g}) - W_{CRDS}(\text{g})}{W_{CRDS}(\text{g}) - W_{HCR}(\text{g})} \times 100\% \quad (7)$$

$$x_s(\%) = (1 - \text{Total porosity (\%)}) \times 100\% \quad (8)$$

$$x_l(\%) = \frac{\text{Initial water content (kg kg}^{-1}\text{)} \times \text{Bulk density (g cm}^{-3}\text{)}}{\rho_{\text{water}}(\text{g cm}^{-3})} \quad (9)$$

$$x_g(\%) = \text{Total porosity (\%)} - \text{Liquid phase (\%)} \quad (10)$$

$$\text{TPSI} = [(x_s - 25)x_l x_g]^{0.4769} \quad (11)$$

Where: the TPSI refers to the three soil phase indices, ranging from 0 to 100. The higher the TPSI value, the better the soil structure.  $x_s$  is the volumetric soil content (%),  $x_l$  is the volumetric water content (%), and  $x_g$  is the volumetric gas content (%) (Wang et al., 2015).

### 2.5 | Statistical analyses

One-way analysis of variance (ANOVA) was performed to assess the physical property differences among the soil depths. Significant differences in the mean root biomass, soil physical properties, and hydrological characteristics between the grazing and enclosure areas were



**TABLE 2** Effect of grazing exclusion on the distribution of plant roots in different soil layers

|                   | Grazing area (kg m <sup>-2</sup> ) | Exclosure area (kg m <sup>-2</sup> ) |
|-------------------|------------------------------------|--------------------------------------|
| 0–10 cm           |                                    |                                      |
| Grass roots       | 0.154 ± 0.014                      | 0.593 ± 0.025***                     |
| Total woody roots | 1.176 ± 0.143                      | 3.318 ± 0.485**                      |
| Fine woody roots  | 1.008 ± 0.121                      | 2.091 ± 0.300**                      |
| 10–20 cm          |                                    |                                      |
| Grass roots       | 0.016 ± 0.005                      | 0.101 ± 0.008***                     |
| Total woody roots | 2.615 ± 0.333                      | 2.463 ± 0.320 ns                     |
| Fine woody roots  | 1.01 ± 0.091                       | 1.365 ± 0.178 ns                     |
| 20–30 cm          |                                    |                                      |
| Grass roots       | 0.004 ± 0.002                      | 0.019 ± 0.006*                       |
| Total woody roots | 67.51 ± 14.002                     | 69.643 ± 11.578 ns                   |
| Fine woody roots  | 0.776 ± 0.139                      | 0.7 ± 0.089 ns                       |
| Total             |                                    |                                      |
| Grass roots       | 0.174 ± 0.015                      | 0.713 ± 0.028***                     |
| Total woody roots | 71.343 ± 13.805                    | 75.331 ± 11.730 ns                   |
| Fine woody roots  | 2.839 ± 0.247                      | 4.069 ± 0.357*                       |

Note: ns, no significant difference.

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

evaluated using the least significant difference test (LSD). A correlation analysis was performed to explore the relationship between soil hydrological and physical properties. Significant correlation coefficients were tested at  $p < 0.05$ ,  $p < 0.01$  and  $p < 0.001$ .

### 3 | RESULTS

#### 3.1 | Root biomass

The root depth distribution of herbs, shrubs, and trees in the grazing and exclosure areas is presented in Table 2. The herb root biomass was significantly higher in the exclosure area than the grazing area, independently to the soil layers (0–10, 10–20, and 20–30 cm) ( $p < 0.001$ ). The total herb root biomass in the exclosure area increased significantly by 307.99% compared with the grazing area in the 0–30 cm soil layer ( $p < 0.001$ ). The total root biomass of shrubs and trees in the exclosure area increased significantly relative to the grazing area in the 0–10 cm layer ( $p < 0.01$ ), whereas no significant differences occurred in 10–20, 20–30, and mean 0–30 cm soil layers. The fine root biomass of shrubs and trees in the exclosure area increased significantly compared with that of the grazing area in the

0–10 cm layer ( $p < 0.01$ ) and in the averaged 0–30 cm layer ( $p < 0.05$ ), whereas no significant differences occurred in the 10–20 and 20–30 cm soil layers. The herb root biomass and the fine root biomass of shrubs and trees in both the grazing and exclosure areas decreased with increasing soil depths. The total root biomass of shrubs and trees increased with the increasing soil depths in the grazing area, while initially, it increased and then decreased in the exclosure area.

#### 3.2 | Soil physical properties and structure

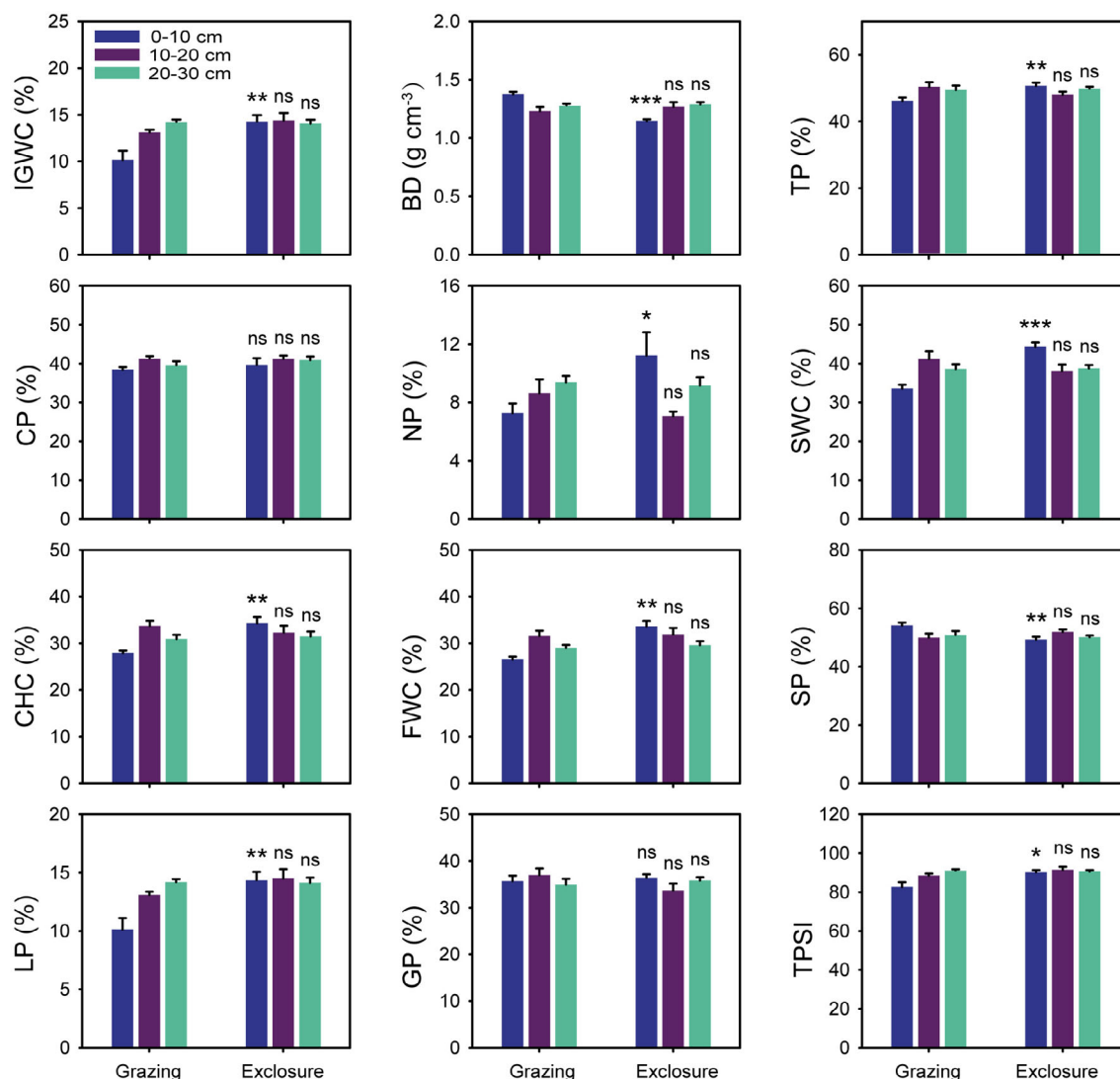
The soil physical property differences indicated soil depth variation between the grazing and exclosure areas (Figure 2). The physical properties of soil water content ( $p < 0.01$ ), total porosity ( $p < 0.01$ ), non-capillary porosity ( $p < 0.05$ ), saturated water capacity ( $p < 0.001$ ), capillary holding capacity ( $p < 0.01$ ), field capacity ( $p < 0.01$ ), and TPSI ( $p < 0.05$ ) in the exclosure area increased significantly, while the soil bulk density ( $p < 0.001$ ) decreased significantly compared with the grazing area in the 0–10 cm soil layer. However, there were no significant differences in the aforementioned soil physical properties between the exclosure and grazing areas in the 10–20 and 20–30 cm soil layers ( $p > 0.05$ ). There were no significant differences in the capillary porosity and gas phase physical properties between the exclosure and grazing areas in either soil layer ( $p > 0.05$ ).

#### 3.3 | Infiltration rate and cumulative infiltration

The cumulative infiltration of water and the corresponding infiltration rate are shown in Figure 3. The cumulative infiltration was consistently higher in the exclosure area than in the grazing area (Figure 3a), with the infiltrated water reaching approximately 20-cm depth within 60 min. However, the infiltrated water exceeded 60-cm depth within 34 min in the exclosure area. The initial infiltration rate of the exclosure area was 4.33 cm min<sup>-1</sup>, which was significantly higher than that of the grazing area (1.20 cm min<sup>-1</sup>,  $p < 0.001$ , Figure 3b). Similarly, the average infiltration rate in the exclosure area increased significantly by 341.77% compared with that of the grazing area ( $p < 0.001$ ).

#### 3.4 | Water flow behaviour

The water flow paths were illustrated directly in the dye-stained vertical section that was excavated in the middle line of the cylinder (Table 3, Figure 4). Preferential flow paths, which resulted from the macropores that consisted of root channels, termite holes, and cracks, were dominant in these two experimental areas. The total dye-stained area in the exclosure area (1975.94 cm<sup>2</sup>) was significantly higher than that in the grazing area (1578.42 cm<sup>2</sup>,  $p < 0.05$ ). The maximum depth and width were 51.67 cm and 54.67 cm, respectively, in the exclosure area, both of which were significantly higher than the corresponding grazing area values of 42.33 cm and 47.67 cm, respectively. In the exclosure and



**FIGURE 2** Physical property differences of each soil layer between the grazing and enclosure areas of the study site. IGWC: initial gravimetric water content (%); BD: bulk density ( $\text{g cm}^{-3}$ ); TP: total porosity (%); CP: capillary porosity (%); NP: noncapillary porosity (%); SWC: saturated water capacity (%); CHC: capillary holding capacity (%); FWC: field water capacity (%); SP: solid phase (%); LP: liquid phase (%); GP: gas phase (%); TPSI: three soil phase index (%). \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ . [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

grazing areas, the percentage of the active water zone ( $>2.0 \text{ g L}^{-1}$ ) in the total stained area was 78.42% and 75.35%, respectively, and the percentage of the water buffer zone ( $<2.0 \text{ g L}^{-1}$ ) was 21.58% and 24.65%, respectively (Table 3). The total stained area and the active water zone initially increased with depth and then decreased.

### 3.5 | Plant community, soil property, and infiltration parameter correlations

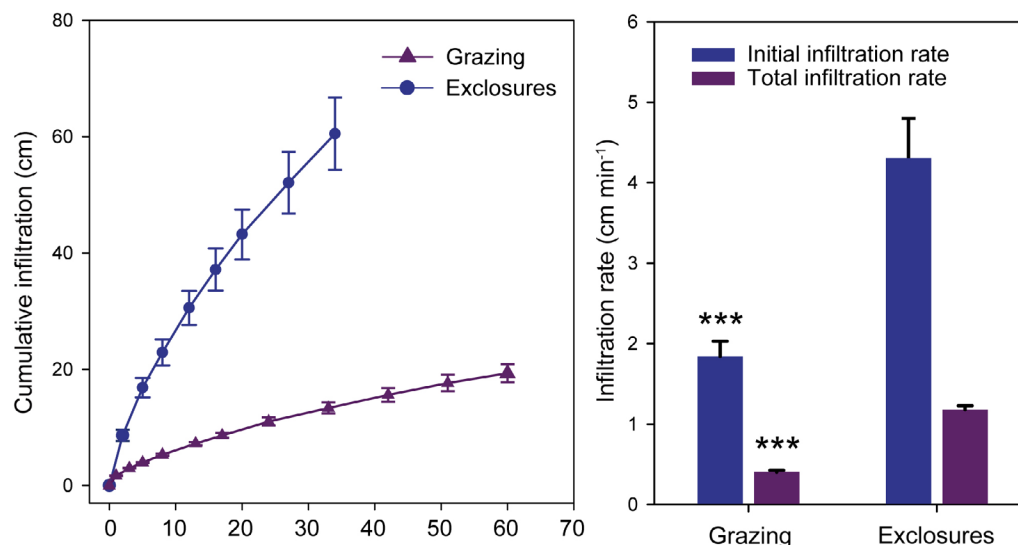
The average infiltration rate, initial infiltration rate, total stained area, and active water zone, as well as the cumulative infiltration, had a significantly positive correlation with herb root biomass ( $p < 0.05$ ), but not with the total and fine root biomass of shrubs and trees ( $p > 0.05$ ; Figure 5). The average infiltration rate, initial infiltration rate, total

stained area, active water zone, and cumulative infiltration had significantly positive correlations with total porosity, saturated water capacity, capillary holding capacity, field capacity, and liquid phase, and significantly negative correlations with bulk density.

## 4 | DISCUSSION

### 4.1 | Enclosure effects on soil physical properties and soil moisture

The results of the present study demonstrated that the enclosure improved the soil's physical properties in the surface soil layer. In a savanna ecosystem, numerous factors may influence soil physical properties, such as grazing intensity, litter cover, roots, geographical



**FIGURE 3** Variation in the (a) cumulative infiltration (cm) and the (b) initial (IIR, cm min<sup>-1</sup>) and average infiltration rates (AIR, cm min<sup>-1</sup>) in the grazing and exclosure areas. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ . [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4388)]

**TABLE 3** Dye stain characteristics in the vertical dye soil profile in the grazing area and exclosure area in the study site

| Treatment      | Infiltration time (min) | Dye-stained width (cm) | Dye-stained depth (cm) | Total stained area (cm <sup>2</sup> ) | Area of concentration (cm <sup>2</sup> ) |                        |
|----------------|-------------------------|------------------------|------------------------|---------------------------------------|------------------------------------------|------------------------|
|                |                         |                        |                        |                                       | >2.0 g L <sup>-1</sup>                   | <2.0 g L <sup>-1</sup> |
| Grazing area   | 360.00                  | 47.67                  | 42.33                  | 1578.42                               | 1189.37                                  | 389.05                 |
| Exclosure area | 26.00**                 | 54.67*                 | 51.67**                | 1975.94**                             | 1549.59**                                | 426.35 ns              |

\* $p < 0.05$ , \*\* $p < 0.01$ .

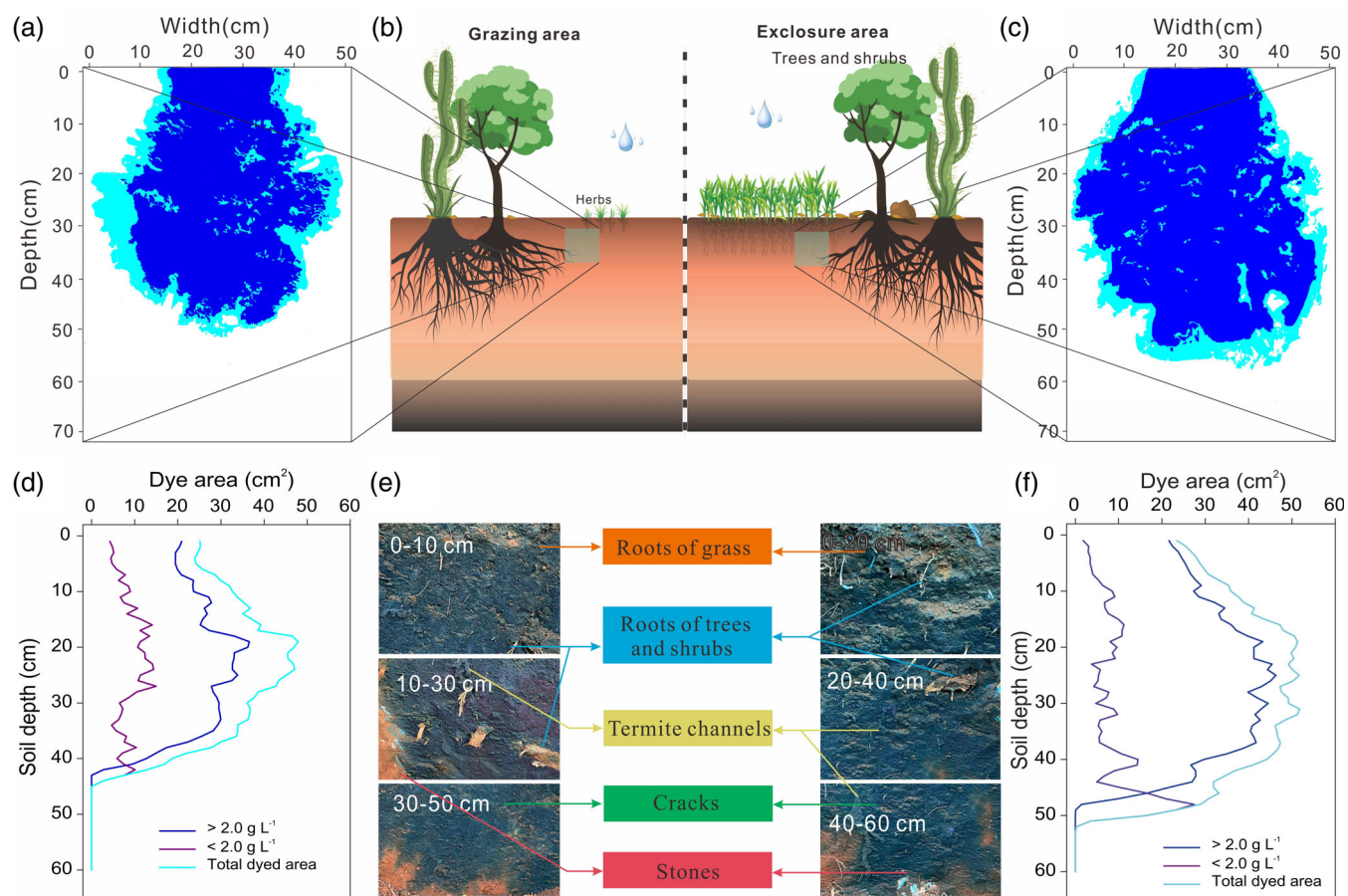
and climatic conditions, and human activities (Savadogo et al., 2007). In this study, the soil bulk density decreased significantly, while the total soil porosity increased significantly in the exclosure area compared with the grazing area (Figure 2b,c). Similar results were obtained in previous studies (Sharow, 2007; Zhou et al., 2010), the researchers mainly attributed these changes to high pressure applied to the soil surface by animal hooves. For example, Houlbrooke et al. (2009) reported a significant improvement in soil density and reduction of soil porosity owing to grazing and trampling by livestock, forcing a 200–350 kPa pressure (produced by sheep and cows) on the surface (Willatt & Pullar, 1984). Capillary porosity refers to the fine capillaries in the soil particles, whereas non-capillary porosity mainly refers to space in-between soil particles (Gupta & Gupta, 1979). In this study, although capillary porosity in the exclosure areas was not significantly different from that of the grazing area, the non-capillary porosity was significantly higher in the former area (Figure 2c,d). These results support the findings of Zhao et al. (2011), who suggested that the topsoil layer in the grazing area suffers from deformation owing to the pressure exerted on the pores and the low non-capillary porosity, with major impacts on water and air conductivity as well as on water retention capacity. In addition, abundant herb roots in exclosure areas could produce more non-capillary porosity by root decay and penetration in the soil matrix (Wu et al., 2016).

The exclosure increased the water content in the surface soil layer (Figure 2a) remarkably. This result corroborates those of Dong et al.

(2015), who revealed that the water content of the soil increased significantly after 11 years of resting from grazing, thereby contributing to the above- and belowground biomass restoration. Vegetation canopies are widely considered to be insulators, which can modify the flux and heat exchange between the soil and the atmosphere and further regulate soil water storage and distribution (Zhao et al., 2011). Thus, the exclosure induced an increase in vegetation and litter cover, reduced the evaporation and radiation exposure, and enhanced water infiltration and the soil water turnover, thereby improving the water content. Additionally, the exclosure increased the root biomass of trees and shrubs in the shallow soil layers (Table 2) and influenced the water cycle positively in soil–plant systems by accelerating the water exchange between the upper and the subsoil layers (Dong et al., 2015). Thus, even when the annual evaporation is significantly greater than the annual precipitation in the Yuanjiang region (southwest China), the water uptake of the shrub and tree roots from the subsoil layers could increase the soil moisture in the topsoil layers.

## 4.2 | Water infiltrability and related influencing factors

Water infiltration is the ability of soil to allow rainfall to travel into the subsurface and ground watertable that determines surface runoff, soil erosion, plant water utilisation, and groundwater recharge (Guo



**FIGURE 4** Dye-stained infiltration patterns for vertical soil sections in (a) the grazing and (c) exclosure areas under (b) the control experiments. Calculated area ( $\text{cm}^2$ ) of the dye-stained soil profile versus depth (cm) of the (d) grazing and (f) exclosure areas. The real structure of different soil layers in the (e) grazing and exclosure areas. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4388)]

et al., 2019; Jiang et al., 2017). In our study, the initial and average infiltration rates in the exclosure area improved significantly compared with those in the grazing area (Figure 3), thereby suggesting that the exclosure improved soil infiltrability dramatically. This result is consistent with those of Wu et al. (2014), who found that soil saturated hydraulic conductivity, a measure of water infiltration, decreased markedly in a steppe grassland of the Loess Plateau after grazing. The soil infiltration increases in the exclosure area likely resulted from the absence of livestock and subsequent 8-year vegetation restoration. Water can hardly infiltrate the soil because of soil compaction caused by the animal hooves in the grazing area, while soil porosity improvements facilitated water infiltration in the exclosure area (Marquart et al., 2019; Savadogo et al., 2007). Additionally, the root systems, especially the herb roots, can act as preferential soil flow paths to increase the downward water flux to deeper soil in the exclosure area (Cheng et al., 2017). Furthermore, the numerous termite holes in the exclosure, coupled with higher organic matter, contributed to soil macropore formation and improved soil structure and hydraulic conductivity (Jiang et al., 2017; Zhu et al., 2019).

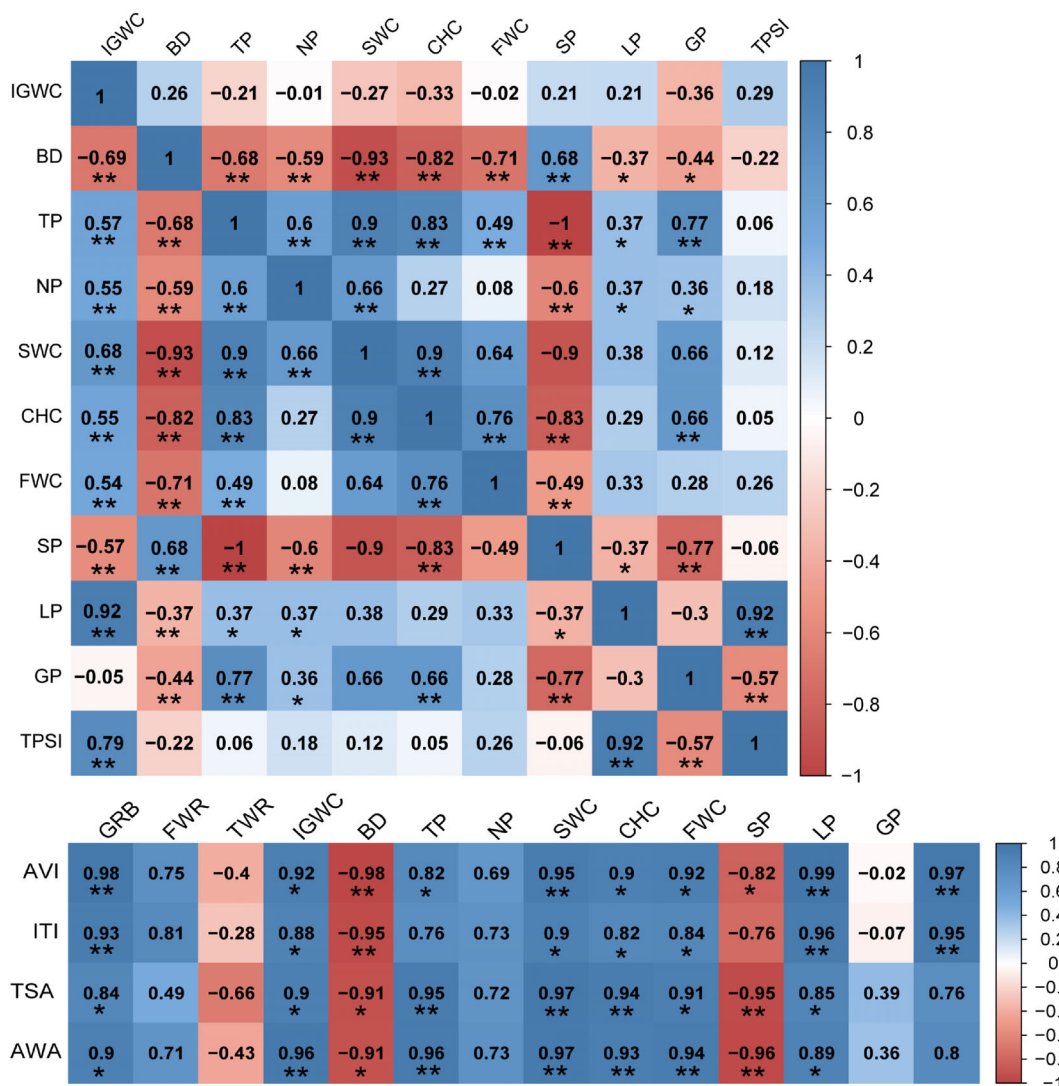
The dye-stained characteristics reveal the degree and pattern of preferential flow and provide visual information on water movement in the soil profiles (Bargués Tobella et al., 2014). The maximum dye-

stained depths and widths in the exclosure area suggested that root systems and animal burrows guided the water and allowed it to infiltrate deeper soil layers and extend the water distribution in the horizontal soil layer (Table 3, Figure 4). The preferential flow that was usually affected by the interaction among soil structures, plant root systems, and soil animals and was higher in the exclosure area than in the grazing area. In our study, the favourable soil structure, dense vegetation cover and root systems, and numerous termite holes in the exclosure area likely promoted preferential flow paths and accelerated water redistribution and availability in vertical and horizontal soil layers, which can diversify the water ingress.

#### 4.3 | Exclosure effect on plant differentiation and related hydrological functions

Grazing can affect the vertical niche differentiation of functional vegetation by altering the distribution of herbaceous and woody plants (shrubs and trees) (Kulmatiski et al., 2010). In agreement with the results of a previous study (Jin et al., 2017), we found that the below-ground biomass of herbs decreased drastically in the grazing area compared with the exclosure area. However, the total root biomass of





**FIGURE 5** Correlation matrices of Spearman's rank correlations among (a) the soil physical properties and (b) among soil water infiltration, movement, and root biomass parameters as well as physical properties. Positive correlations are displayed in blue and negative correlations in red colour. The colour intensity of the blocks is proportional to the correlation coefficients. In the right side of the correlogram, the legend colour shows the correlation coefficients and the corresponding colours. IGWC: initial gravimetric water content (%); BD: bulk density ( $\text{g cm}^{-3}$ ); TP: total porosity (%); CP: capillary porosity (%); NP: noncapillary porosity (%); SWC: saturated water capacity (%); CHC: capillary holding capacity (%); FWC: field water capacity (%); SP: solid phase (%); LP: liquid phase (%); GP: gas phase (%); TPSI: three soil phase index (%); AIR: average infiltration rate; IIR: initial infiltration rate; TSA: total stained area; AWA: The active water zone area; GRB: grass root biomass; FWR: fine woody root biomass; TWR: total woody root biomass. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ . [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4388)]

trees and shrubs remained unaffected by grazing, probably because livestock species preferentially consume herbs and avoid shrubs and trees. Moreover, the fine and total roots increased in the enclosure area compared with the grazing area in the 0–10 cm soil layer; however, there was no difference between the two areas in the 10–30 cm soil layer (Table 2), which could be attributed to a significant improvement in the physicochemical properties and water content in the shallow layers, but no significant changes in the deep layers (Figure 2 and Table 1).

The change in niche differentiation among plants in the grazing and enclosure areas could result in variations in hydrological functions and services. A previous study found that trees (but not herbs) appear to be highly plastic in terms of their water use, using surface soil water

during seasons with abundant rainfall and deep soil water (below 30 cm in depth) during the dry season (Holdo & Nippert, 2015). This suggests that the woody roots are sensitive to soil water, with water-controlled distribution patterns resulting in higher and lower root biomass when the water content is high and low, respectively (Wu et al., 2019). We also found a significant positive correlation between the biomass of fine and total tree roots and soil moisture (Figure 5), which corroborates the plasticity of woody plant roots in their uptake patterns. Additionally, the herb plant roots mainly absorb water from the surface soil layer in grazing and enclosure areas because of their morphometric features and shallow roots (Holdo & Nippert, 2015; Priyadarshini et al., 2016). However, woody plants absorbed water mainly from the deep soil layers in the grazing area, while in the

enclosure area they absorbed water from both the shallow and the deep soil layers. Thus, the different water uptake patterns of woody plants in the enclosure area can reduce the interspecific (e.g. between shrubs and trees) plant competition, while at the same time increasing the competition between herb and woody plants (Barron-Gafford et al., 2017; Wu et al., 2019). However, long woody roots can transport water to shallow soil layers; then, they absorb water from deeper soil layers, which is added to surface soil layers. In addition, the dense herbaceous cover in the enclosure area likely increased the water content of the shallow layers by enhancing water infiltration during the rainy season and acting as an insulator to reduce evaporation in the dry season. Thus, despite the interspecific competition for water uptake, the coexistence of herb and woody plants promoted water availability in the shallow soil layers. Therefore, the enclosures that induce niche differentiation between the roots of herbs and woody species may act in a complementary manner in terms of water capture and exchange between different soil layers.

Our results support the hypothesis suggesting tree-grass differences in terms of functional rooting distribution in the present study, and suggest that herb and woody plants play different roles in soil infiltration and water movement. Studies have shown that herb plants can generate more preferential root channels in shallow soil layers (Jiang et al., 2017). The herb roots connected surface soil to deeper soil layers in the enclosure area and acted as a conducting route, as we observed during the dye tracing experiment, thereby resulting in a preferential water flow. This was likely a major reason for the higher initial infiltration rate in the shallow layer in the enclosure area. However, the roots of woody plants were mainly concentrated in the deeper soil layers (Table 2), where they played a vital role in the vertical water infiltration of the deep soil layers and acted as lateral water flow paths in all layers.

#### 4.4 | Implications for grassland management

There is a consensus that overgrazing affects soil physical and hydrological parameters of grassland (Pulido et al., 2018; Teague et al., 2011). Previous review of worldwide studies [excluding the data from Hoshino et al. (2009)] demonstrated that livestock grazing increases soil bulk density by 0.08–15.87% and decreased infiltration rate by 9.02–89.28% in both grasslands and savanna systems (Table S1). Simultaneously, some studies suggest that grazing exclusion could improve the soil structures and hydrological characteristics in savanna and steppe systems. In addition, grazing exclusion also restores the vegetation cover and soil nutrients and reduces soil erosion, thereby preventing landscape degradation (Fernández-Lugo et al., 2009; Feyisa et al., 2017; Koci et al., 2020). Besides grazing exclusion, proper grazing strategies such as continuous light and alternate grazing with low or moderate grazing intensity could also improve soil quality and fertility in the pasture land (Abdalla et al., 2018; Teague et al., 2011). For example, Teague et al. (2011) indicated that multi-paddock grazing is superior for conserving and restoring resources (e.g., vegetation, soil nutrients, and soil

productivity) than light and heavy continuous grazing in north Texas tall grass prairie. Furthermore, environmental conditions and grassland type also influence soil quality in grazing land. A previous review indicated that the grassland management policies at different locations should consider the local climate and pasture types (Abdalla et al., 2018). In Yuanjiang (China), numerous livestock (mostly cows and goats) are raised under the local smallholders' range condition, resulting in a continuous harmful impact on the local environment. Therefore, other grazing strategies such as alternate grazing, continuous light grazing, and multi-paddock grazing should also been proposed and tested by the researcher considering soil and water conservation, environment protection, and sustainable economic development.

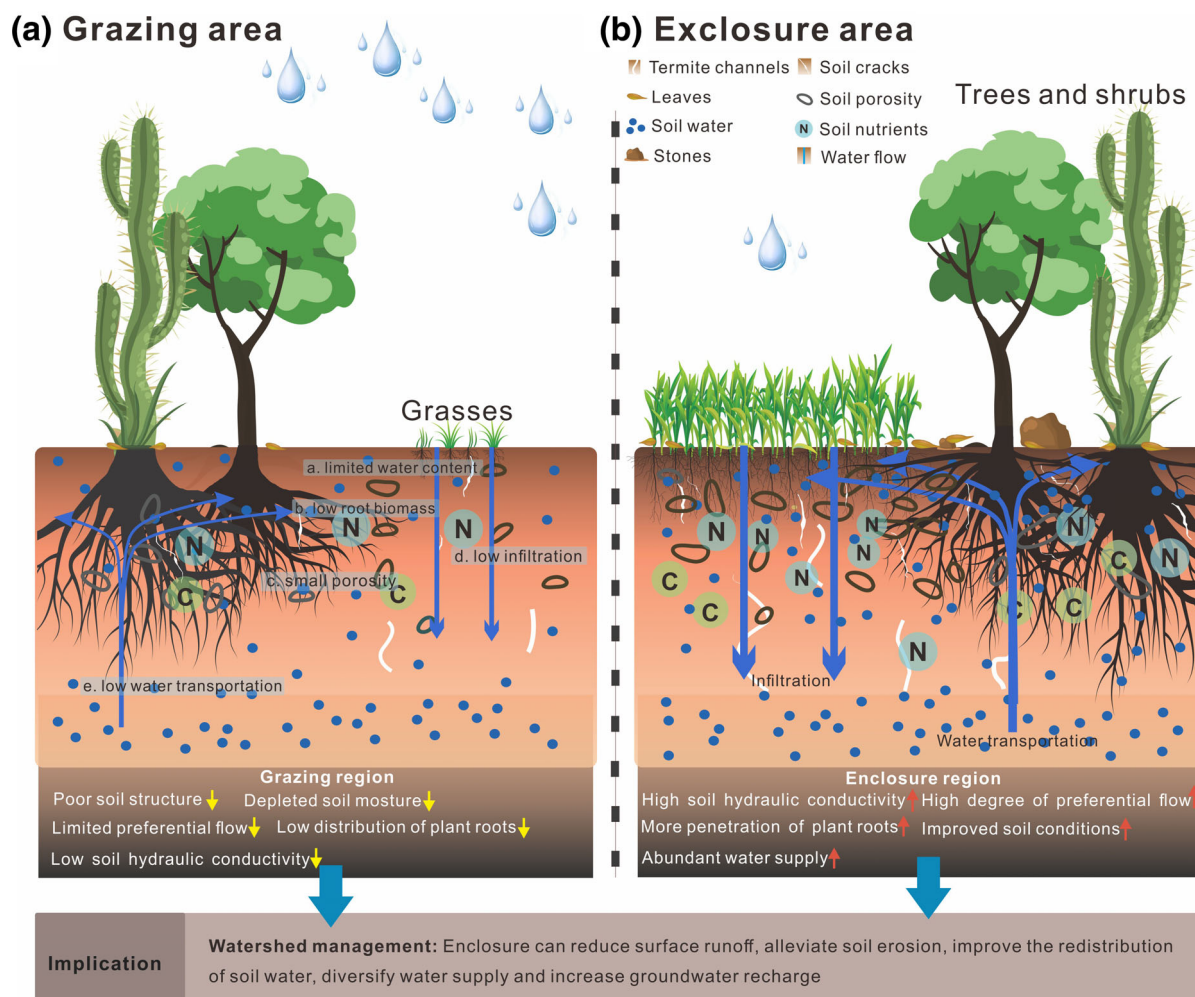
#### 4.5 | Limitations of the research

Previous study using stable isotope-based has demonstrated the different water use patterns between trees and grasses with rooting differences in savannah systems (Holdo & Nippert, 2015). Given that the present study merely focused on the soil hydrological processes (e.g., the movement, redistribution, and recharge of water), the knowledge of different plant communities' water use efficiency is lacking. Therefore, the future investigation should focus on the assessment of water movement and supply in different soil profiles and their subsequent influence on water use efficiency of different plant communities in the present savannah systems.

Although this study has demonstrated that enclosure could improve the soil hydrological functions (e.g., increase in soil infiltrability and preferential flow paths) and soil organic matter and total N (Table 1), it is probably not suitable for grazing management at large scale considering about local economic development (Mekuria et al., 2011). In addition, besides the soil hydrological characteristics, vegetation composition and coverage, soil microflora and nutrients, and soil water availability were also closely related with soil quality (Mureithi et al., 2014; Verdoodt et al., 2010). Thus, more control experiments with different grazing strategies and animal stocking density at a large scale should been designed and established to understand their environmental, ecological, and economical benefits in the savanna systems.

### 5 | CONCLUSIONS

This study examined the enclosure effects on hydrological functions and services, including water distribution, exchange, infiltration, and flow behaviour in a savanna ecosystem. The results demonstrate a significant improvement in soil hydrologic properties due to an increase in soil infiltrability and preferential flow paths in the enclosure compared with the grazing area. The enclosure area was also characterised by improved soil physical properties, such as lower bulk density and higher porosity, field water capacity, and TPSI. Combined with the increased root and faunal activities in the enclosure area, these soil



**FIGURE 6** Schematic diagram illustrating the enclosure effect on hydrologic processes and functions in a livestock grazing area in a savannah system. The soils in the grazing area suffered from (a) limited soil moisture and (b, c) poor soil structure and (d) hydrological conditions. However, enclosures can significantly improve the hydrological functions through improving the soil structure and restoration and the niche differentiation of vegetation. The enclosure not only improved significantly the soil physical properties (increased bulk density, decreased total and noncapillary porosities), but also increased the plant root biomass of herb and woody plants. More herb plants can accelerate water infiltration and reduce evaporation, while woody plants can facilitate water exchange in different vertical and horizontal soil layers and transport the groundwater to the shallow layer, thereby supplementing the surface soil water in the enclosure area. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

properties enhanced the infiltration capacity and promoted hydraulic redistribution in the three-dimensional soil profiles. The enclosure area was characterised by significantly increased root biomass of herb and woody plants in the shallow layer, while this niche differentiation contributed to different hydrological functions. The herbaceous species promoted water infiltration into deep soil layers and reduced evaporation loss by insulating the surface soil in the enclosure area. However, the presence of woody species accelerated water exchange between different vertical soil layers and improved lateral preferential water flow paths, thereby supplying water from deeper soil layers or groundwater to the surface layer (Figure 6). These results indicate that the enclosure-induced restoration of herb and woody species could lead to the redistribution of water resources and the diversification of water supply, thereby promoting the water availability for different plant species. Thereby, these findings confirmed that enclosures play

essential roles in ecohydrological processes. They can increase water infiltration and redistribution, accelerate recharge, and attenuate soil erosion, thereby contributing to soil water conservation and management in the savanna. As enclosure can naturally recover vegetation, improve soil structures, promote water recharge, availability, and storage, it should be appropriately expanded in the present savanna systems even when prioritising economic interest over sustainable development.

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## DATA AVAILABILITY STATEMENT

The data which supports findings of this study are available from the corresponding author upon a reasonable request.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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